Lecture 3: Eigenvalue Problems

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1 Linear Algebra

Eigenvalues and Eigenvectors: Let $A \in \mathbb{R}^{n\times n}$ be a matrix and $x \in \mathbb{R}^n/\{0\}$ be a non-zero vector. We say x is an eigenvector of A and $\lambda \in \mathbb{R}$ is its corresponding eigenvalue if

$$Ax = \lambda x \tag{1}$$

Eq (1) is equivalent to

$$(\lambda I - A)x = 0$$

which is a system of linear equations. In order to guarantee that non-zero solution of x exists, we must have $\text{rank}(\lambda I - A) < n \Leftrightarrow \det(\lambda I - A) = 0$.

Geometric Multiplicity: For any eigenvalue λ, the set of all the x that give $(\lambda I - A)x = 0$ is the eigenspace of λ, denoted as E_λ. E_λ includes all the eigenvectors corresponding to λ together with the zero vector. In fact, $E_\lambda = \text{null}(\lambda I - A)$. We say $\dim(E_\lambda)$ is the geometric multiplicity of λ.

Theorem 1: Let λ_1 and λ_2 be two eigenvalues of A. If $\lambda_1 \neq \lambda_2$, then $E_{\lambda_1} \cap E_{\lambda_2} = \{0\}$.
Proof: We prove the contrapositive statement: If $E_{\lambda_1} \cap E_{\lambda_2} \neq \{0\}$, then $\lambda_1 = \lambda_2$. First, it is impossible for $E_{\lambda_1} \cap E_{\lambda_2} = \varnothing$ because $x = 0$ is a common solution of $(\lambda I - A)x = 0$ for any λ. If $E_{\lambda_1} \cap E_{\lambda_2} \neq \{0\}$, then there must be an non-zero vector $x \in E_{\lambda_1} \cap E_{\lambda_2}$. Since $x \in E_{\lambda_1}$, we have $Ax = \lambda_1 x$. Since $x \in E_{\lambda_2}$, we have $Ax = \lambda_2 x$. Thus $\lambda_1 x = \lambda_2 x$. Since x is non-zero, we must have $\lambda_1 = \lambda_2$.

Note that $E_{\lambda_1} \cap E_{\lambda_2} = \{0\}$ implies the basis of E_{λ_1} and the basis of E_{λ_2} are linearly independent.

Characteristic Polynomial: The characteristic polynomial of A is defined as $p_A(z) = \det(zI - A)$. If λ is an eigenvalue of A, then $p_A(\lambda) = 0$.

Algebraic Multiplicity: Let $\lambda_1, \lambda_2, ..., \lambda_n$ be the roots of $p_A(z) = 0$, then $p_A(z) = (z - \lambda_1)(z - \lambda_2)...(z - \lambda_n)$. $\lambda_1, \lambda_2, ..., \lambda_n$ are the eigenvalues of A. An eigenvalue may appear more than once. We say the multiplicity of λ as a root of p_A is the algebraic multiplicity of λ.

Similarity Transformation: Let $X \in \mathbb{R}^{n\times n}$ be a non-singular matrix, then the map $A \to X^{-1}AX$ is called the similarity transformation of A.

Theorem 2: If X is non-singular, then A and $X^{-1}AX$ have the same characteristic polynomial, eigenvalues, and algebraic and geometric multiplicities.
Proof: The equivalence of characteristic polynomial can be shown as follows:

$$\begin{aligned} p_{X^{-1}AX}(z) &= \det(zI - X^{-1}AX) = \det(X^{-1}(zI - A)X) = \det(X^{-1})\det(zI - A)\det(X) \\ &= \det(zI - A) = p_A(z) \end{aligned}$$

Since the eigenvalues and algebraic multiplicity are uniquely determined by characteristic polynomial, A and $X^{-1}AX$ have the same eigenvalues and algebraic multiplicity.

Now we discuss the equivalence of geometric multiplicity. Note that if $x \in \mathbb{R}^n$ is the solution of $(\lambda I - A)x = 0$, then let $y = X^{-1}x$, $(\lambda I - X^{-1}AX)y = X^{-1}(\lambda I - A)Xy = 0$, and vice versa. This implies that if E_λ is an eigenspace for A, then for the same λ, $X^{-1}E_\lambda$ is the eigenspace for $X^{-1}AX$. Thus we have $\dim(E_\lambda) = \dim(X^{-1}E_\lambda)$.

Theorem 3: The algebraic multiplicity of an eigenvalue λ is at least as great as its geometric multiplicity.
Proof: Suppose the geometric multiplicity of λ for the matrix A is k where $k \leq n$, we construct an orthogonal matrix V where the first columns of V are the orthogonal basis of E_λ, then

$$B = V^T AV = \begin{bmatrix} \lambda I & C \\ 0 & D \end{bmatrix}$$

where I is $k \times k$ identity matrix, C is $k \times (n-k)$, D is $(n-k) \times (n-k)$. Since $A \to B$ is a similarity transformation, A and B will have the same eigenvalues and algebraic multiplicity. Since

$$\begin{aligned} \det(zI - B) &= \det\left(\begin{bmatrix} (z-\lambda)I & -C \\ 0 & zI - D \end{bmatrix}\right) = \det((z-\lambda)I)\det(zI - D) \\ &= (z-\lambda)^k \det(zI - D) \end{aligned}$$

This implies the algebraic multiplicity of λ is at least k, because the term $\det(zI - D)$ may contain the factor $z - \lambda$.

Defective Eigenvalues and Matrices: An eigenvalue whose algebraic multiplicity exceeds its geometric multiplicity is a defective eigenvalue. A matrix that has one or more defective eigenvalues is a defective matrix.
Example: Consider two matrices

$$A = \begin{bmatrix} 2 & & \\ & 2 & \\ & & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 & \\ & 2 & 1 \\ & & 2 \end{bmatrix}$$

Both A and B has only one eigenvalue $\lambda = 2$ with algebraic multiplicity 3. For A, the basis of E_λ can be e_1, e_2, e_3, so the geometric multiplicity is 3. For B, there is only one vector e_1 in the basis of E_λ, so the geometric multiplicity is 1. Thus, A is non-defective, B is defective.

Eigenvalue Decomposition: In Eq (1), suppose we get the eigenpairs $(\lambda_1, x_1), (\lambda_2, x_2)$, $...(\lambda_n, x_n)$, Let $X = [x_1, x_2, ..., x_n]$ and $\Lambda = \mathrm{diag}(\lambda_1, \lambda_2, ..., \lambda_n)$. Then

$$AX = X\Lambda$$

If X is non-singular, it has an inverse X^{-1}. Then we get

$$A = X\Lambda X^{-1} \tag{2}$$

We call Eq (2) the eigenvalue decomposition of A.

Theorem 4: A matrix $A \in \mathbb{R}^{n\times n}$ is non-defective if and only if it has an eigenvalue decomposition $A = X\Lambda X^{-1}$.
Proof: ($\Longleftarrow$): If $A = X\Lambda X^{-1}$ exists, then $\Lambda \to A$ is a similar transformation. By Theorem 2, Λ and A will have the same eigenvalues, algebraic multiplicity and geometric multiplicity. Since Λ is a diagonal matrix, it is non-defective, which implies A is non-defective.

($\Longrightarrow$): If A is non-defective, then for any eigenvalue of A, the geometric multiplicity will be equal to algebraic multiplicity. By Theorem 1, A must have n linearly independent eigenvectors. Thus X is non-singular, $A = X\Lambda X^{-1}$ exists.

2 Eigenvalue Decomposition for Symmetric Matrix

Theorem 5: If $A \in \mathbb{R}^{n\times n}$ is a symmetric matrix, then there exists orthogonal matrix $Q \in \mathbb{R}^{n\times n}$ and diagonal matrix Λ such that

$$A = Q\Lambda Q^T \tag{3}$$

Eq (3) is the eigenvalue decomposition of A, this implies that a symmetric matrix always has an eigenvalue decomposition, thus symmetric matrix is non-defective.

The proof of Theorem 5 will use Schur Factorization. If A can be decomposed as $A = QTQ^T$ where Q is orthogonal and T is upper-triangular, we call the decomposition as Schur factorization.

Theorem 6: Every square matrix A has a Schur factorization.
Proof: we prove by the induction on the dimension n of A. $n = 1$ case obviously holds. When $n \geq 2$, we assume any $(n-1)\times(n-1)$ square matrix has a Schur factorization, and show that the $n \times n$ matrix will also have a Schur factorization.

Let $A \in \mathbb{R}^{n\times n}$ and (λ, u_1) be an eigenpair of A where we let u_1 be a unit vector. We generate a orthogonal matrix $U \in \mathbb{R}^{n\times n}$ where u_1 is the first column of U. Remember that all other columns of U can be generated by Gram-Schmidt orthogonalization. Then

$$AU = U\begin{bmatrix}\lambda & B\\ 0 & C\end{bmatrix} \quad\Longrightarrow\quad A = U\begin{bmatrix}\lambda & B\\ 0 & C\end{bmatrix}U^T$$

where C is an $(n-1)\times(n-1)$ matrix. By our assumption, C has a Schur factorization $C = VTV^T$ where V is orthogonal and T is upper triangular. Then

$$A = U\begin{bmatrix}\lambda & B\\ 0 & VTV^T\end{bmatrix}U^T \quad\Longrightarrow\quad A = U\begin{bmatrix}1 & 0\\ 0 & V\end{bmatrix}\begin{bmatrix}\lambda & BV\\ 0 & T\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & V^T\end{bmatrix}U^T$$

Let $Q = U\begin{bmatrix}1 & 0\\ 0 & V\end{bmatrix}$, then Q is orthogonal, we get

$$A = Q\begin{bmatrix}\lambda & BV\\ 0 & T\end{bmatrix}Q^T$$

Since $\begin{bmatrix}\lambda & BV\\ 0 & T\end{bmatrix}$ is upper-triangular, $A = Q\begin{bmatrix}\lambda & BV\\ 0 & T\end{bmatrix}Q^T$ is a Schur factorization.

□

Since any $A \in \mathbb{R}^{n\times n}$ has a Schur factorization $A = QTQ^T$, when A is symmetric, we have $A = A^T \Rightarrow QTQ^T = QT^TQ^T \Rightarrow T = T^T$. This implies T is a diagonal matrix. Thus we proved Theorem 5.

Differences between Eigenvalue Decomposition and SVD: Eigenvalue decomposition and singular value decomposition (SVD) are two different types of matrix decomposition.

- Any matrix, either squared or non-squared, has an SVD. Only squared non-defective matrix has eigenvalue decomposition.
- For SVD $A = U\Sigma V^T$, U and V are orthogonal, and we may not have $U = V$ even when they are of the same size. The diagonal elements of Σ (singular values) are non-negative. For eigenvalue decomposition $A = X\Lambda X^{-1}$, X may not be orthogonal, and the diagonal elements of Λ (eigenvalues) can be either negative or non-negative.

Relationships between Eigenvalue Decomposition and SVD for Symmetric Matrix:

Theorem 7: If $A \in \mathbb{R}^{n\times n}$ is symmetric, and λ is an eigenvalue of A, then $|\lambda|$ is a singular value of A.

Proof: Suppose A has an SVD $A = U\Sigma V^T$. Since $A = A^T$, we have $A^2 = AA^T = U\Sigma V^T V\Sigma U^T = U\Sigma^2 U^T$. Note that $A^2 = U\Sigma^2 U^T$ is both an SVD and an eigenvalue decomposition.

Since A has an eigenvalue decomposition $A = Q\Lambda Q^T$, then $A^2 = (Q\Lambda Q^T)(Q\Lambda Q^T) = Q\Lambda^2 Q^T$ is also an eigenvalue decomposition.

Since the eigenvalue decomposition is not unique, we may not have $U = Q$. Note that $A^2 \to \Sigma^2$ and $A^2 \to \Lambda^2$ are both similarity transformation of A^2, which means Σ^2 and Λ^2 have the same eigenvalues. Suppose the eigenvalues on the diagonals of Σ^2 and Λ^2 are both arranged in descending order, then $\Sigma^2 = \Lambda^2 \Rightarrow \Sigma = |\Lambda|$. This implies that if λ is an eigenvalue in Λ, then $|\lambda|$ must be a singular value in Σ.

□

Example: Consider we have a symmetric matrix $A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$, then A has an eigenvalue decomposition

$$\begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 0.81 & -0.57 & 0.12 \\ 0.57 & 0.71 & -0.42 \\ 0.15 & 0.40 & 0.90 \end{bmatrix} \begin{bmatrix} 2.70 & & \\ & 0.76 & \\ & & -1.46 \end{bmatrix} \begin{bmatrix} 0.81 & 0.57 & 0.15 \\ -0.57 & 0.71 & 0.40 \\ 0.12 & -0.42 & 0.90 \end{bmatrix} \tag{4}$$

If we move the negative sign of the eigenvalue -1.46 to either the left or the right orthogonal matrix:

$$\begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 0.81 & -0.57 & -0.12 \\ 0.57 & 0.71 & 0.42 \\ 0.15 & 0.40 & -0.90 \end{bmatrix} \begin{bmatrix} 2.70 & & \\ & 0.76 & \\ & & 1.46 \end{bmatrix} \begin{bmatrix} 0.81 & 0.57 & 0.15 \\ -0.57 & 0.71 & 0.40 \\ 0.12 & -0.42 & 0.90 \end{bmatrix} \tag{5}$$

or

$$\begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 0.81 & -0.57 & 0.12 \\ 0.57 & 0.71 & -0.42 \\ 0.15 & 0.40 & 0.90 \end{bmatrix} \begin{bmatrix} 2.70 & & \\ & 0.76 & \\ & & 1.46 \end{bmatrix} \begin{bmatrix} 0.81 & 0.57 & 0.15 \\ -0.57 & 0.71 & 0.40 \\ -0.12 & 0.42 & -0.90 \end{bmatrix} \tag{6}$$

then both Eq (5) and Eq (6) will be the SVDs of A.

Theorem 8: If $A \in \mathbb{R}^{n\times n}$ is positive semi-definite, then all eigenvalues of A are non-negative. Consequently, there exists an orthogonal matrix Q and diagonal matrix Λ such that $A = Q\Lambda Q^T$ is both an eigenvalue decomposition and an SVD.

Proof: Since A is symmetric, A has an eigenvalue decomposition $A = Q\Lambda Q^T$. Since A is positive semi-definite, for any $x \in \mathbb{R}^n$, $x^T Ax \geq 0$. Let $y = Q^T x$, we have

$$x^T Ax = x^T Q\Lambda Q^T x = y^T \Lambda y = \sum_{i=1}^{n} \lambda_i y_i^2 \geq 0$$

This implies that $\lambda_i \geq 0$ for all i. Thus all elements on the diagonal of Λ are non-negative, $A = Q\Lambda Q^T$ is also an SVD.

□

3 Spectral Theorems

Rayleigh Quotient: Let $A \in \mathbb{R}^{n\times n}$ be a symmetric matrix, and $x \in \mathbb{R}^n$, then the Rayleigh quotient of x is defined as

$$r_A(x) = \frac{x^T A x}{x^T x}$$

The Rayleigh quotient of an eigenvector is its eigenvalue, i.e., if $Ax = \lambda x$, then

$$r_A(x) = \frac{x^T A x}{x^T x} = \frac{x^T(\lambda x)}{x^T x} = \lambda \frac{x^T x}{x^T x} = \lambda$$

Courant-Fischer Theorem:
Theorem 9 (Courant-Fischer): Let $A \in \mathbb{R}^{n\times n}$ be a symmetric matrix with eigenvalues $\lambda_1 \geq \lambda_2 \geq ... \geq \lambda_n$, and $x \in \mathbb{R}^n$ be a vector, then for any $k \in \{1, 2, ..., n\}$,

$$\lambda_k = \max_{\substack{S\subseteq\mathbb{R}^n \\ \dim(S)=k}} \min_{\substack{x\in S \\ x\neq 0}} \frac{x^T A x}{x^T x} = \min_{\substack{T\subseteq\mathbb{R}^n \\ \dim(T)=n-k+1}} \max_{\substack{x\in T \\ x\neq 0}} \frac{x^T A x}{x^T x}$$

Proof: We show the first equality as follows. Consider the two statements:
(1) $\exists S \subseteq \mathbb{R}^n, \dim(S) = k$ such that $\min_{x\in S, x\neq 0} \frac{x^T A x}{x^T x} \geq \lambda_k$.
(2) $\forall S \subseteq \mathbb{R}^n, \dim(S) = k$, we have $\min_{x\in S, x\neq 0} \frac{x^T A x}{x^T x} \leq \lambda_k$.
If both (1) and (2) holds, then

$$\lambda_k = \max_{\substack{S\subseteq\mathbb{R}^n \\ \dim(S)=k}} \min_{\substack{x\in S \\ x\neq 0}} \frac{x^T A x}{x^T x}$$

.

Let's prove (1) first. Since A is symmetric, A must have an eigenvalue decomposition $A = Q\Lambda Q^T$, where Q is orthogonal. Let q_i be the ith column of Q, and suppose q_i is the eigenvector of the eigenvalue λ_i.

Take $S = \text{span}(q_1, q_2, ..., q_k)$. For any $x \in S$ and $x \neq 0$, we can write $x = \sum_{i=1}^k c_i q_i$ where $c_i \in \mathbb{R}$ and $\sum_{i=1}^k c_i^2 > 0$. Then

$$\begin{aligned}\frac{x^T A x}{x^T x} &= \frac{(\sum_{i=1}^k c_i q_i)^T (\sum_{j=1}^k c_j A q_j)}{(\sum_{i=1}^k c_i q_i)^T (\sum_{j=1}^k c_j q_j)} = \frac{(\sum_{i=1}^k c_i q_i)^T (\sum_{j=1}^k c_j \lambda_j q_j)}{(\sum_{i=1}^k c_i q_i)^T (\sum_{j=1}^k c_j q_j)} \\ &= \frac{\sum_{i=1}^k \sum_{j=1}^k c_i c_j \lambda_j q_i^T q_j}{\sum_{i=1}^k \sum_{j=1}^k c_i c_j q_i^T q_j} = \frac{\sum_{i=1}^k \lambda_i c_i^2}{\sum_{i=1}^k c_i^2} \geq \frac{\sum_{i=1}^k \lambda_k c_i^2}{\sum_{i=1}^k c_i^2} = \lambda_k\end{aligned}$$

This implies

$$\min_{\substack{x\in S \\ x\neq 0}} \frac{x^T A x}{x^T x} \geq \lambda_k$$

Now we prove (2). For any $S \subseteq \mathbb{R}^n, \dim(S) = k$, let $T \subseteq \mathbb{R}^n, T = \text{span}(q_k, q_{k+1}, ..., q_n)$, then $\dim(T) = n-k+1$. Since $\dim(S)+\dim(T) > n$, we must have $\dim(S\cap T) \geq 1$. Suppose there exists a nonzero vector $x \in S \cap T$, then we can write x as $x = \sum_{i=k}^n c_i q_i$ where $c_i \in \mathbb{R}$ and $\sum_{i=k}^n c_i^2 > 0$. Then

$$\frac{x^T A x}{x^T x} = \frac{\sum_{i=k}^n \lambda_i c_i^2}{\sum_{i=k}^n c_i^2} \leq \lambda_k$$

Since for any nonzero vector $x \in S$, there exists $x \in S \cap T$ that satisfies the above inequality, we have

$$\min_{\substack{x \in S \\ x \neq 0}} \frac{x^T A x}{x^T x} \leq \lambda_k$$

The second equality can be proved similarly by replacing A with $-A$.

$\square$

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